IX.

On a Problem Concerning Division.

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In a paper[2] belonging to the year 1849, it was incidentally observed that, in dividing an integer n by all divisors not larger than it, the case in which the remainder lies below half the divisor occurs more often than the opposite case, in which the remainder exceeds or equals half the divisor. It was also shown there that, as n increases, the ratio of the number of divisors for which the first case occurs to their total number n tends to the limit $2 - \log 4 = 0.61370\ldots$. It seems of some interest to generalize the investigation and to determine the number h of those divisors $1, 2, \ldots, p$, where $p \leq n$, for which the corresponding remainder has ratio to the divisor less than a given proper fraction α. If square brackets are used to denote the greatest integer contained in the enclosed value, so that $x - [x]$ is always zero or a positive proper fraction, then it is easy to see that the divisor s will or will not have the required property according as the difference

$$\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]$$

is equal to the positive unit or to zero. Thus one has

$$h = \sum_1^p \left(\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]\right) = \sum_1^p \left[\frac{n}{s}\right] - \sum_1^p \left[\frac{n}{s} - \alpha\right],$$

where, here and throughout what follows, the summation sign refers to s. In this form the expression for h is neither well suited to numerical computation, nor does it show how h changes for increasing values of n and p. A form serving this double purpose is obtained by the following transformation, which is also applicable in many other cases.

Let $y = f(x)$ be a function which continually decreases as the variable x increases from $x = \mu$ to $x = p$. The inverse function $x = F(y)$ will plainly have the same character and will likewise continually decrease, while the variable y increases from $y = f(p)$ to $y = f(\mu)$. Taking the constants μ and p to be integers, put, for abbreviation,

$$[f(\mu)] = \nu, \qquad [f(p)] = q,$$

and form the sequence

$$[f(\mu)],\ [f(\mu+1)],\ \ldots,\ [f(s)],\ \ldots,\ [f(p)],$$

[1] In the 1851 volume of the Academy reports the communication is introduced on p. 20 with the words: Herr Lejeune Dirichlet laid before the class the following note on a problem concerning the theory of division. In the printing on pp. 151–154 of volume 47 of Crelle's Journal (1854), Dirichlet altered the title and text somewhat. I have here used this later version almost throughout. K.

[2] On the determination of mean values in number theory. Vol. II, p. 49 of this edition of G. Lejeune Dirichlet's Works. The cited remark is found on p. 57. K.

in which each term is equal to or exceeds the following one. We shall now determine which terms of this sequence are equal to an arbitrary integer t lying between q and ν.

For this purpose first seek the completely determined index s of the term whose value is $\geq t$, while the following term is $< t$. Thus one has

$$[f(s)] \geq t, \qquad [f(s+1)] < t,$$

or, what is the same thing,

$$f(s) \geq t, \qquad f(s+1) < t,$$

and from the hypothesis made on the function $f(x)$ it follows that

$$s \leq F(t), \qquad s+1 > F(t),$$

hence

$$s = [F(t)]\,.$$

Applying this result to t and $t+1$, one sees that the value t belongs only to those terms whose index s satisfies the double condition

$$s > [F(t+1)]\,, \qquad s \leq [F(t)]\,.$$

Because the beginning and end of the sequence are prescribed, this result is modified for $t = \nu$ by requiring the first condition to be $s \geq \mu$, and for $t = q$ by replacing the second condition by $s \leq p$.

Taking this result into account, it is now easy to transform the sum

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s),$$

where $\varphi(s)$ denotes an entirely arbitrary function: first combine all terms for which $[f(s)]$ has one and the same value, and then add all the partial sums so obtained. Put

$$\sum_{1}^{s} \varphi(s) = \Psi(s).$$

Then the partial sum in which $[f(s)]$ has the value t is

$$\begin{array}{ll} t\big(\Psi[F(t)] - \Psi[F(t+1)]\big), & \text{if } q < t < \nu, \\ \nu\big(\Psi[F(\nu)] - \Psi(\mu)\big), & \text{if } t = \nu, \\ q\big(\Psi(p) - \Psi[F(q+1)]\big), & \text{if } t = q, \end{array}$$

and therefore

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s) = q\Psi(p) - \nu\Psi(\mu) + \sum_{q+1}^{\nu} \Psi[F(s)].$$

Now separate, in each of the two sums contained in the expression for h given above, the first μ terms, and apply the formula just found to the remaining terms. This gives

$$\sum_{\mu+1}^{p} \left[\frac{n}{s}\right] = \sum_{q+1}^{\nu} \left[\frac{n}{s}\right] + pq - \mu\nu,$$

where $\left[\frac{n}{\mu}\right] = \nu$ and $\left[\frac{n}{p}\right] = q$, and

$$\sum_{\mu+1}^{p} \left[\frac{n}{s} - \alpha\right] = \sum_{q'+1}^{\nu'} \left[\frac{n}{s+\alpha}\right] + pq' - \mu\nu',$$

where $\left[\frac{n}{\mu}-\alpha\right]=\nu'$ and $\left[\frac{n}{p}-\alpha\right]=q'$.

For abbreviation denote $\nu-\nu'$ by δ and $q-q'$ by ε, so that δ and ε can have only the values 0 or 1. Put the last sum in the form

$$\sum_{q'+1}^{\nu'}\left[\frac{n}{s+\alpha}\right]=\sum_{q+1}^{\nu}\left[\frac{n}{s+\alpha}\right]+\varepsilon\left[\frac{n}{q+\alpha}\right]-\delta\left[\frac{n}{\nu+\alpha}\right],$$

and substitute. One obtains

$$\begin{aligned}h=\sum_{1}^{\mu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s}-\alpha\right]\right)+\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)\\ +\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon-\left(\mu-\left[\frac{n}{\nu+\alpha}\right]\right)\delta.\end{aligned}$$

Although this transformation is valid for all values of p, it is advantageous only when p is greater than $\sqrt{n}$. Under this hypothesis it is most advantageous when, as will be done below, one chooses for the number μ, which has so far been arbitrary, one of the integers adjacent to $\sqrt{n}$. As is easy to see, the number of divisions needed for the exact determination of h is then approximately

$$2\sqrt{n}-\frac{n}{p},$$

whereas the original expression required p divisions.

We shall now, under the hypothesis that p is of higher order than $\sqrt{n}$, that is, that $\frac{p}{\sqrt{n}}$ grows beyond every bound with n, seek to determine the limiting value of the ratio $\frac{h}{p}$ of the number of divisors having the required property to their total number p. In this investigation one may neglect, in the expression for h, all terms whose order is lower than that of p. Dropping the first, whose order does not exceed $\sqrt{n}$, and also the fourth, which can contain only a bounded number of units, one obtains

$$h=\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)+\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon,$$

or also, if one omits the square brackets, which plainly causes only a change whose order does not exceed $\sqrt{n}$,

$$h=n\sum_{q+1}^{\nu}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\varepsilon.$$

If the upper limit ν is changed to ∞, the sum receives the increment

$$\frac{1}{\nu+1}-\frac{1}{\nu+1+\alpha}+\frac{1}{\nu+2}-\frac{1}{\nu+2+\alpha}+\cdots<\frac{1}{\nu+1},$$

whose value, multiplied by n, likewise does not exceed the order $\sqrt{n}$. Thus one obtains

$$\lim\frac{h}{p}=\frac{n}{p}\sum_{q+1}^{\infty}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\frac{\varepsilon}{p}.$$

It remains to distinguish the case where the quotient $\frac{n}{p}$, which by hypothesis is ≥ 1, grows beyond every bound, from the case where this quotient remains finite.

In the first case the second term approaches zero, while the ratio of the sum contained in the first term to $\frac{\alpha}{q}$ tends to unity. Thus the limit of $\frac{h}{p}$ coincides with that of $\frac{n}{pq}\alpha$, that is, with α.

In the second case, where $\frac{n}{p}$, and hence also $q = \left[\frac{n}{p}\right]$, remains finite, it is expedient to bring the limiting value of $\frac{h}{p}$ given directly by the last equation into another form. Instead of the sum one introduces the difference of two others, taken from $s = 1$ to respectively $s = \infty$ and $s = q$, and then expresses the first by an integral. Our equation then becomes

$$\lim \frac{h}{p} = \frac{n}{p} \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi - \frac{n}{p} \sum_1^q \left(\frac{1}{s} - \frac{1}{s + \alpha}\right) + \left(1 - \frac{n}{p} \frac{1}{q + \alpha}\right) \varepsilon,$$

where the integral, which for every rational value of α can be represented by logarithms and circular functions, is a known transcendental function studied many times. If one sets in particular $p = n$, then $q = 1$, $q' = 0$, $\varepsilon = 1$, and the limiting expression passes into

$$\lim \frac{h}{n} = \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi.$$

With the help of the table of these transcendents given by Gauss in the paper *Disquisitiones generales circa seriem infinitam* ...[3], one can easily determine the value of α corresponding to a given value of the integral. For example, one finds that, if for half the divisors $1, 2, \dots, n$ the ratio of the remainder to the divisor is to lie below α, then $\alpha = 0.384686\dots$ is required.

[3]Gauss' Works, vol. III, p. 161. K.